

Week 2 HW Question 4:

Date:

$$x_1 - 2x_2 + 3x_3 = 24$$

$$-x_1 + 3x_2 - x_3 = -11$$

$$2x_1 - 5x_2 + 5x_3 = 42$$

$$\begin{bmatrix} 1 & -2 & 3 \\ -1 & 3 & -1 \\ 2 & -5 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 24 \\ -11 \\ 42 \end{bmatrix}$$

$$\begin{array}{l} R_1 + R_2 \rightarrow R_2 \\ -2R_1 + R_3 \rightarrow R_3 \end{array} \sim \begin{bmatrix} 1 & -2 & 3 & | & 24 \\ 0 & 1 & 2 & | & 13 \\ 0 & -1 & -1 & | & -6 \end{bmatrix}$$

$$R_2 + R_3 \rightarrow R_3 \sim \begin{bmatrix} 1 & -2 & 3 & | & 24 \\ 0 & 1 & 2 & | & 13 \\ 0 & 0 & 1 & | & 7 \end{bmatrix}$$

$$\begin{array}{l} -2R_3 + R_2 \rightarrow R_2 \\ -3R_3 + R_1 \rightarrow R_1 \end{array} \sim \begin{bmatrix} 1 & -2 & 0 & | & 3 \\ 0 & 1 & 0 & | & -1 \\ 0 & 0 & 1 & | & 7 \end{bmatrix}$$

$$2R_2 + R_1 \rightarrow R_1 \sim \begin{bmatrix} 1 & 0 & 0 & | & 1 \\ 0 & 1 & 0 & | & -1 \\ 0 & 0 & 1 & | & 7 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 7 \end{bmatrix}$$